Unit: Unit 1: Ratio, Rate, Percentage
Topic: Ratios Proportions

Name: ________________
Date: ____________

Foundation (Jalapeno)

Q1. A basketball team scored 3 points for every 5 points their opponents scored. What is the ratio of the team's points to their opponents' points?
(A) 3 : 5
(B) 5 : 3
(C) 2 : 3
(D) 3 : 2
(E) 1 : 1

Q2. A cyclist travels 24 miles in 2 hours. If she maintains the same rate, how many miles will she travel in 5 hours?
(A) 48 miles
(B) 60 miles
(C) 72 miles
(D) 84 miles
(E) 96 miles

Q3. A sports drink contains 12 grams of carbohydrates per 100 milliliters. What is the ratio of carbohydrates to total volume?
(A) 1 : 8
(B) 2 : 17
(C) 3 : 25
(D) 12 : 100
(E) 1 : 10

Q4. A swimmer completes 8 laps in 20 minutes. If she maintains the same rate, how many laps will she complete in 30 minutes?
(A) 10 laps
(B) 12 laps
(C) 15 laps
(D) 18 laps
(E) 20 laps

Q5. A tennis player wins 3 out of every 5 games. What is the ratio of games won to total games played?
(A) 3 : 5
(B) 5 : 3
(C) 2 : 3
(D) 3 : 2
(E) 1 : 1

Q6. A runner completes 4 miles in 30 minutes. If she maintains the same rate, how many miles will she complete in 45 minutes?
(A) 5 miles
(B) 6 miles
(C) 7 miles
(D) 8 miles
(E) 9 miles

Q7. A volleyball team scores 18 points per game, while their opponents score 12 points per game. What is the ratio of the team's points to their opponents' points?
(A) 3 : 2
(B) 2 : 3
(C) 1 : 1
(D) 4 : 3
(E) 3 : 1

Q8. A gymnast performs 6 routines in 18 minutes. If she maintains the same rate, how many routines will she perform in 24 minutes?
(A) 7 routines
(B) 8 routines
(C) 9 routines
(D) 10 routines
(E) 11 routines

Open Ended Questions (FRQ)

Q9. A baseball player hits 2 home runs for every 7 games played. If he plays 35 games, how many home runs will he hit? Show your work and explain your reasoning.

Q10. A cyclist travels 50 miles in 4 hours. If she maintains the same rate, how many miles will she travel in 6 hours? Use proportions to solve the problem and show your work.

Chef's Tips

- Ensure students show their work and explain their reasoning for free-response questions.
- Encourage students to use proportions to solve problems.
- Remind students to check their units and ensure they are consistent throughout the problem.